

Security of the Internet of Space Things (IoST)

An Overview of IoST Security
Protocols and Attack
Methodologies

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CONTENTS

01 What Is IoST?

03 Meet the SatOver
Attack

05 Simple Shields Up

02 Why Satellites Can
Be Hacked

04 Real-World Fallout

06 Your Next Steps

Internet of Space Things in One Minute

Imagine your phone ordering pizza, but instead of using Wi-Fi or cell towers, it chats directly with a passing satellite . That's the Internet of Space Things (IoST) – everyday gadgets using satellites for a truly global connection.

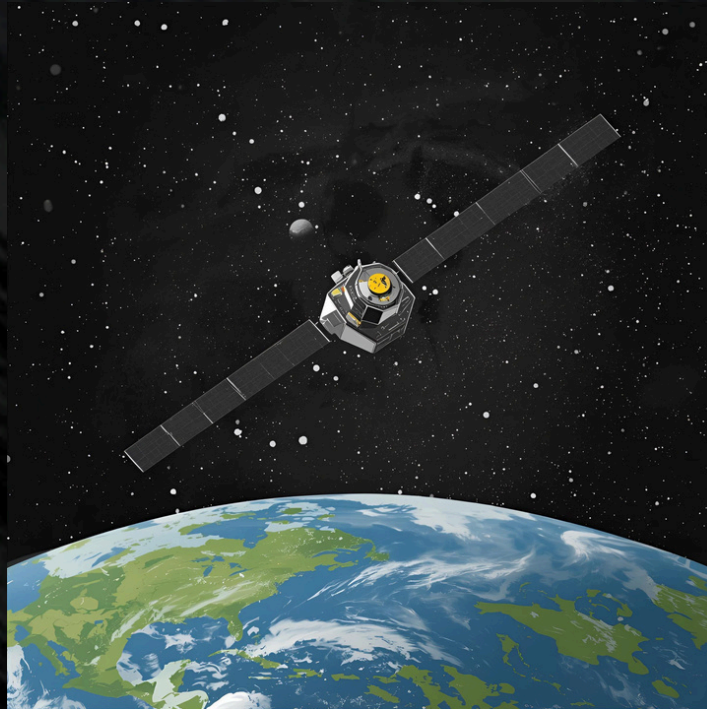
It's like having a direct line to space, making the world even more connected.



What Is IoST?

Tiny Satellites, Big Job

Meet the CubeSats: small, fast, and essential.



Shoebox-Sized Helpers

These tiny satellites, called CubeSats, orbit Earth every 90 minutes.



A Swarm in the Sky

They work together in a "constellation," passing your message like relay runners.

Their small size makes them cheap and global, but also tricky to protect.

Why Satellites Can Be Hacked

Open Doors in Space

Why satellites can be easy targets for hackers.



Familiar Signals

Many reuse signals like those from cell phones, so your device can connect easily.



Unencrypted Hello

The initial connection isn't encrypted, making it vulnerable to eavesdropping.



Easy to Drown Out

An attacker can simply "shout" louder with a stronger signal to interfere.

It's like someone cutting in line at a coffee shop because no one checked their order.

Why Satellites Can Be Hacked

The Congestion Chaos Trick

A simple hack to create a massive traffic jam in the sky.



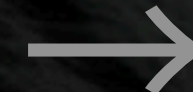
1. Attacker Sends Fake Signal

A simple "I'm full" message is sent to a ground station.



2. Ground Station Trusts It

The station believes the fake message and thinks the satellite is overloaded.



3. Real Users Are Blocked

The station slows down or blocks real users, causing an outage.

This attack locks the sky for minutes without ever triggering a security alarm.

The SatOver Attack: A Play-by-Play

Meet the clever bully that floods the network with fake busy signals.

The Attacker's Move

-  Pretends to be hundreds of fast-moving satellites.
-  Floods the network with fake "congestion" messages.
-  Hides by constantly changing its identity.

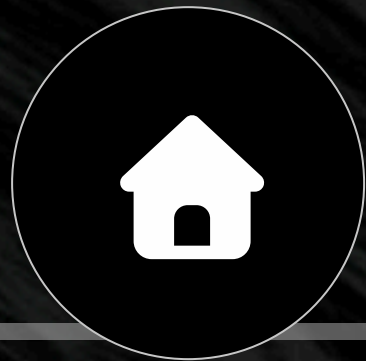
The Victim's Experience

-  Device keeps retrying to connect, getting nowhere.
-  Experiences a complete denial of service.

Defending against this is like a frustrating game of whack-a-mole.

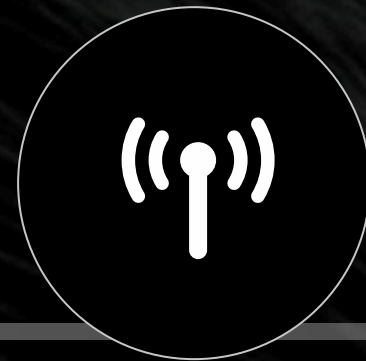
From One City to the Whole Sky

How a single attack can scale to block an entire constellation.



1. Rooftop Antenna

The attack starts with a single, powerful antenna.



2. Hijack Cell Towers

The attacker uses normal cell towers as amplifiers.



3. City-Wide Attack

Each tower becomes a source of fake satellite signals.



4. Whole Sky Blocked

The entire constellation appears unusable to users.

When Phones Can't Call Home

A satellite hack has real-world consequences that hit close to home.



Transportation Grinds to a Halt

Ride-share and delivery apps fail, leaving people and packages stranded.



Farmers Lose Critical Data

Precision agriculture systems stop working, risking crop health and yields.



Hikers Can't Call for Help

Emergency beacons and satellite phones are useless in a crisis.

This isn't science fiction; it's a direct hit to our economy and safety within minutes.

Who Profits from the Mess?

The motives behind satellite hacking are often plain old greed.



Land-Based Carriers

Traditional phone companies could block satellites to force users back onto their own paid networks, eliminating competition.



Criminals & Extortionists

Hackers can demand "protection" fees from satellite companies or users to stop the attacks, creating a lucrative blackmail scheme.

Quick Wins for Device Makers

Simple software fixes can make devices much more resilient.



Add a Watchdog Timer

Program devices to ignore repeated "congestion" messages after a certain threshold. This prevents them from getting stuck in a retry loop.



Smarter Satellite Switching

Teach gadgets to quickly jump to a different satellite if they detect a problem, rather than waiting for a timeout.

These are tiny code tweaks that significantly reduce an attack's impact without needing new hardware.

Smart Network Tricks

Ground stations can get smarter to detect and defeat these attacks.



Anomaly Detection

Ground stations can monitor for sudden, unusual spikes in "congestion" reports from a single geographic area, which is a clear sign of a spoofing attack.



Time-of-Arrival Checks

By measuring when a signal arrives (like a GPS game of Marco Polo), the network can verify if a signal is coming from a real satellite in orbit or a fake source on the ground.

Your Next Steps

Stay Curious, Stay Safe

What you can do to protect yourself and help secure the IoST.



Update Your Firmware

Always install updates for your IoT devices. These often contain important security patches.



Use Strong Passwords

Secure your devices and home network with strong, unique passwords to prevent unauthorized access.



Ask Questions

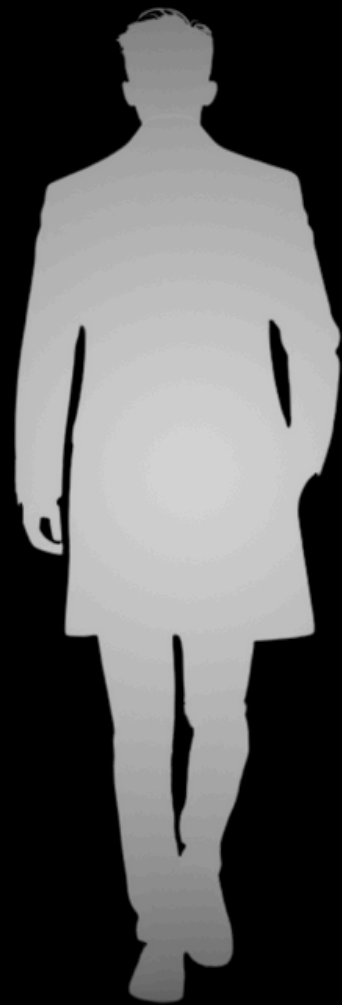
Ask vendors if their devices verify satellite IDs. Demand better security from the products you buy.

Awareness is the cheapest and most effective firewall against future sky hacks.

Space Is Everyone's Yard

We all share the responsibility for keeping space safe and usable. Just like a shared park, if we pick up the trash (bad code) and lock the gates (use strong protocols), everyone can enjoy a safer, more connected world under starry skies.

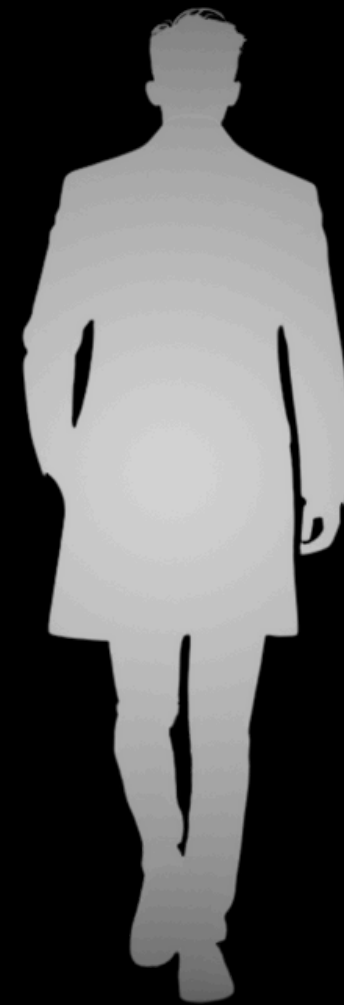




Thank You



Regards @MajorKali



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